Indications and Usefulness of Common Injections for Nontraumatic Orthopedic Complaints

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KEYWORDS

• Joint injections • Steroid injections • Corticosteroids • Tendinopathy • Arthritis

KEY POINTS

- Corticosteroid injections (CSIs) are commonly used in the treatment of painful musculoskeletal conditions, despite a lack of consensus in the literature of the true usefulness.
- Any benefits of CSIs are of modest magnitude and short lived, on the order of a few weeks. There are no long-term benefits, and no change in future need for surgical intervention.
- Hyaluronic acid injections to treat knee osteoarthritis are widely used, although the benefits are modest and short term, and the cost is high.
- Injections for treatment of painful musculoskeletal conditions are generally safe and well tolerated, although in some circumstances there are suggestions of long-term deleterious outcomes.

INTRODUCTION

Pain related to various musculoskeletal (MS) conditions is a common patient complaint, and one that is often difficult to remedy. In addition to oral analgesics and physical therapy, local injections (most commonly of corticosteroid [CS]) are a common intervention and have been for decades. However, in most cases, the literature is full of poor-quality studies, making the true utility of these injections questionable. This article reviews some of the literature studying these injections with the goal of providing clinicians the information to make evidence-based, high-value choices.

OSTEOARTHRITIS OF THE KNEE

Osteoarthritis of the knee is a common problem in Western societies, especially in the elderly, and is one of the biggest causes of disability.[1] There is no cure, and joint

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replacement surgery is expensive and carries significant risks. Therefore, treatment is focused on pain relief interventions and maintaining function. Injection therapy (notably CSs and hyaluronic acid derivatives) are widely used. Various other modalities have been studied in this regard, including exercise therapy, braces, and oral medications, and these must be kept in context for comparison purposes.

Corticosteroid Injections for Osteoarthritis of the Knee: Efficacy

Corticosteroid injections (CSIs) have been used in osteoarthritis of the knee since the 1950s, and there have been numerous studies, many of them small and with significant methodological flaws that limit the interpretation of the results. For example, many studies did not use validated pain scales, and frequently the severity of osteoarthritis was not reported. The studies that are the most useful are double blinded and placebo controlled, and use validated measures of improvement, such as validated pain scales and objective functional measures.

A meta-analysis by Arroll and Goodyear-Smith[2] in 2004 found 10 studies of reasonably good quality and determined that intra-articular steroid injections led to statistically significant improvements in pain, stiffness, and function at 1 week and 2 weeks but not at various later end points, up to 6 months. When they combined the 2 studies that followed patients for 24 weeks, they found a statistically significant improvement compared with baseline, but no difference compared with placebo. The meta-analysis also suggested that higher doses of steroids, such as 40 mg of methylprednisolone or 40 mg of triamcinolone, are superior to the lower doses used in some studies.

A later systematic review was done by Hepper and colleagues[3] in 2009, in which 6 randomized placebo-controlled studies were examined. They used a 100-point analog pain scale, and all 6 of these studies showed a significant benefit from steroid injection at both 1 and 2 weeks, with a reduction in pain of about 50% (20–33 points on the scale). Note that 3 of 5 studies also showed a benefit to placebo at these time points, but to a smaller scale (7%–20%, up to 20 points on the analog scale). All studies showed that the steroid injection was superior in efficacy to the placebo, which was usually saline and/or lidocaine. Similar to the Arroll and Goodyear-Smith[2] meta-analysis, the incremental benefit of the steroid injections seemed to fade after 2-weeks.

Bannuru and colleagues[4] in 2015 did a network meta-analysis and systematic review to compare the efficacy of various interventions in treating osteoarthritis of the knee. They concluded that intra-articular CSIs were superior to oral placebo, injected placebo, and oral nonsteroidal antiinflammatory medications in the short term. They also found that injected placebo was superior to oral placebo. However, they commented on the wide variety of studies and the inconsistency in the literature, and stated that further studies were warranted.

The aforementioned studies were all limited to the effects of 1 injection, but it is possible that repeated injections may have cumulative benefits over time. Raynauld and colleagues[5] studied 68 patients with moderate osteoarthritis of the knee and randomized them in double-blind fashion to receive either triamcinolone 40 mg or saline every 3 months for 2 years (total of 8 injections). They found borderline significant benefit to the steroid injections at 1 year in night pain, as well as a general trend toward improvement in many other measures, but these differences over time were small and did not reach statistical significance in this small study. At 2 years, there were no differences and all trends had disappeared.

A recent Cochrane systematic review[6] concluded that the data for intra-articular steroid injection utility in the treatment of osteoarthritis remains overall very poor, but that there is probably a moderate benefit of corticosteroid injection for 2 weeks, a small benefit at 8 weeks, and little or no benefit at 12 to 26 weeks.

Corticosteroid Injections for Osteoarthritis of the Knee: Safety

In general, intra-articular steroid injections of the knee are very low risk. Typical side effects of CSs, such as transient worsening of diabetes control, are mild and self-limited. Local reactions such as pain, redness, and swelling may occur. Very rarely injection can lead to infections. One concern that is often discussed is possible hastening of cartilage wasting and worsening of osteoarthritis as a result of repeated steroid injections. In the Raynauld and colleagues[5] study mentioned earlier, patients received a total of 8 injections over the course of 2 years, and no deleterious effects were noted between the placebo saline and CSI groups. This study included an analysis of fluoroscopically determined joint space.

Corticosteroid Injections for Osteoarthritis of the Knee: Cost

$50 to $200 per injection, including injection charge and medication.

Corticosteroid Injections for Osteoarthritis of the Knee: Conclusions

Despite the variable results and overall poor data quality, it seems that CSIs are modestly helpful in some patients with knee osteoarthritis in the short term, but do not seem to be helpful in the long term. They are generally safe and well tolerated, and, as long as the patient is clear on the expectations, may help in the management of pain and dysfunction in patients with osteoarthritis of the knee. This advice is in keeping with the American College of Rheumatology 2012 guidelines,[7] although the American Academy of Orthopedic Surgeons 2013 guideline could not recommend for or against the use of CSIs for treatment of osteoarthritis of the knee because of inconclusive evidence.[8,9]

Intra-articular Injection of Hyaluronic Acids in Knee Osteoarthritis: Efficacy

Hyaluronic acid is a naturally occurring substance found in synovial fluid, and is thought to act as a lubricant and shock absorber. Several preparations of varying molecular weights have been developed for intra-articular injection in patients with osteoarthritis of the knee, and are usually termed viscosupplementation.[1] When assessing efficacy, there are many conflicting results and many studies that lacked proper placebo groups or had other methodological flaws. Some studies found no effect on pain and function, whereas others found a positive effect, including one that had a surprisingly large reduction in pain on visual analog scales, with a reduction of 30 points on a 100-point scale. To address these discrepancies, several systematic reviews and meta-analyses have been performed.

Arrich and colleagues[10] found 22 articles that included quantification of pain scales, and on systematic review and meta-analysis there was a mean reduction in visual analog score pain scales of 3 to 7 points (out of 100) at various time points out to 6 months. There was high heterogeneity among trials, and many did not include intention-to-treat analysis, or failed to have blinding. They concluded that the aggregate data are flawed overall, preventing firm conclusions, but there may be a small benefit, of questionable clinical significance, to hyaluronic acid injections. Rutjes and colleagues[11] did a more recent systematic review and meta-analysis, and found 71 trials that showed a benefit to hyaluronate injections. However, they also uncovered unpublished studies that did not show benefit, and concluded that, overall, based on the available data, there is a small clinically insignificant benefit.

In contrast, a Cochrane Review in 2006 concluded that “viscosupplementation is an effective treatment for OA [osteoarthritis] of the knee with beneficial effects: on pain, function and patient global assessment; and at different post injection periods but especially at the 5 to 13 week post injection period.”[12] The review pointed out the wide variability in responses seen in the different studies and the use of different products. The investigators were not able to comment on the benefits of one product compared with another except that a trend existed favoring the higher-molecular-weight substances compared with lower-weight substances.

The American Academy of Orthopedic Surgery revised its guidelines in 2013, stating that intra-articular hyaluronic acid is no longer recommended as a method of treatment of patients with symptomatic osteoarthritis of the knee. They cited a lack of clinically meaningful benefits in meta-analysis.

Intra-articular Injection of Hyaluronic Acids in Knee Osteoarthritis: Safety

There are no studies showing any long-term risks with intra-articular injections of hyaluronic acids, although this has been poorly studied. The injections have a statistically significant increased risk of local adverse effects compared with placebo (Relative Risk = 2), primarily in the form of flares of osteoarthritis that are thought to be caused by inflammatory reaction to the local injection. In some patients this is severe and function limiting, but reversible.

Intra-articular Injection of Hyaluronic Acids in Knee Osteoarthritis: Cost

The cost of these injections can vary widely depending on the product selected and the physician charge. In addition, depending on the product, the number needed can range from 1 to 5 injections. However, costs seem to be between $1200 and $1500.

Intra-articular Injection of Hyaluronic Acids in Knee Osteoarthritis: Conclusion

The efficacy of hyaluronates in the treatment of osteoarthritis remains in question, although most studies suggest either no clinically significant benefit or a modest benefit, without major harms, but at significant cost. These injections may be used, but only in specific circumstances after other, more proven treatments have been used.

SHOULDER PAIN (IMPINGEMENT SYNDROME)

CSIs in patients with shoulder pain have been used for more than 50 years. However, the data on their efficacy are conflicting and variable. Many studies are poorly designed, lacking validated pain scales and placebo controls. Another major problem with the literature on this subject is the wide variety of causes of shoulder pain, including rotator cuff tears, rotator cuff tendinopathy, acromioclavicular joint arthritis, glenohumeral joint arthritis, subacromial bursitis, and adhesive capsulitis. Many studies do not specify the cause, or do not show how the underlying cause was determined. To complicate matters further, the presence of a finding on imaging does not necessarily prove the cause of the problem, because many patients have asymptomatic rotator cuff disease or shoulder arthritis. This article focuses on the use of CSI for treatment of the common impingement syndrome conditions, including bursitis and rotator cuff tendinopathy.

Corticosteroid Injection for Shoulder Impingement Syndrome: Efficacy

Koester and colleagues[13] did a systematic review of the literature as of 2007, assessing the efficacy of subacromial injections in the treatment of rotator cuff disease. They found 9 randomized placebo-controlled studies that specifically studied patients with rotator cuff disease, either tendinopathy alone or with partial cuff tears. All of the studies had small numbers of subjects, all less than 100. Some of the studies included only patients with short periods of symptoms (several weeks), and others studied patients with chronic (>6 months) pain. All of the studies measured pain, either on visual analog scales or by other means, as well as range of motion. Of the 9 studies, only 4 found a statistically significant reduction in pain scales (up to 24 weeks), and 3 showed an improvement in range of motion. However, the clinical significance of these changes was questionable overall, especially with range-of-motion improvements, which tended to be small (range, 5°–10°). The investigators concluded that, overall, the benefit of CSIs in rotator cuff disease is not supported by the literature.

In contrast, Arroll and Goodyear-Smith[14] did a meta-analysis of CSIs for painful shoulders and found a benefit to subacromial injection overall, potentially out to 9 months. This analysis used a dichotomous variable of improvement versus no improvement, and calculated a number needed to treat (NNT) of 2 to 3 overall. However, this review did not limit the type of shoulder disorder, and there was a high degree of heterogeneity among the studies (which places these conclusions in significant doubt overall).

In one of the better designed trials, Alvarez and colleagues[15] compared subacromial injections of betamethasone plus xylocaine with xylocaine alone (control group) in 58 patients with up to 6 months of symptoms related to rotator cuff tendinopathy or partial cuff tear. Of note, these patients were randomized and blinded, and had already failed a more conservative approach of nonsteroidal antiinflammatory medications plus physical therapy. At various time points out to 6 months, a modest improvement in symptoms occurred in both groups, but there was no significant difference between the two groups overall in shoulder function, pain, or quality-of-life measures.

In addition, in a recent study Rhon and colleagues[16] compared subacromial CSIs (up to 3 in total, spaced 1 month apart) with 3 weeks of manual physical therapy in 100 randomized patients with shoulder impingement syndrome. Both groups had a significant 50% improvement in SPADI (shoulder pain and disability index) scores compared with baseline at all points in the study (1–12 months). However, the two groups did not differ in efficacy, and the injection group had more health care interactions, and twice as many subjects returned throughout the year for injection compared with the physical therapy group. However, both groups had improvement compared with baseline, and the investigators concluded that both of these interventions offer benefit to patients with impingement syndrome.

Corticosteroid Injection for Shoulder Impingement Syndrome: Safety

In the various randomized studies, no significant complications were reported apart from local pain and redness. Specifically, there were no reports of tendon rupture. There are case reports of septic bursitis, usually with *Staphylococcus aureus*, following subacromial injection,[17] but this is a rare complication. Worsening of glycemic control may occur in patients given CSIs, although generally this is a self-limited issue. More importantly, concerns have been raised about the effects of CSIs on the rotator cuff tendons, and various animal studies have shown significant softening and other changes to the tendons with repeat injections. One study found a higher rate of cuff repair failure in patients who had more than 3 subacromial CSIs preoperatively, although this has not been replicated elsewhere and is not definitive.[18]

Corticosteroid Injection for Shoulder Impingement Syndrome: Cost

Total cost for CSI of the shoulder (medication and physician fee) can vary, but generally ranges from $50 to $200.

Corticosteroid Injection for Shoulder Impingement Syndrome: Conclusions

In patients with impingement syndrome, including subacromial bursitis and rotator cuff tendinopathy or partial tears, CSI seems to have limited benefit that is not necessarily clinically significant for most patients. On the whole, CSI does not seem to be superior to physical therapy in the long term, although individual responses vary. There are no common significant risks. CSI is likely to remain an option for treatment of impingement syndrome, especially in patients with severe pain or who have not responded to other measures.

LATERAL EPICONDYLITIS (EPICONDYLALGIA)

Lateral epicondylitis is a common condition characterized by pain at the lateral epicondyle of the elbow, caused by abnormalities in the wrist extensor muscles. There is debate as to whether there is an inflammatory component to this condition, thus some clinicians prefer the term lateral epicondylalgia. Different interventions exist for this condition, including physical therapy/stretching, braces, CSI, and rarely surgery. This article focuses on the utility of CSI.

Corticosteroid Injection for Lateral Epicondylalgia Syndrome: Efficacy

Many of the studies in the literature are poor in quality, but there have been some reasonably well-designed placebo-controlled studies from which some conclusions can be drawn.

A 2002 study by Smidt and colleagues[19] randomized 185 patients to 1 of 3 groups: CSI, physiotherapy, or wait and see (no treatment). There was no placebo injection group, but the therapist who rated the patients' symptoms was blinded to the treatment allocation. At the 6-week mark, the injection group had significantly higher success rates than the other two groups: 92% success versus 47% in the physiotherapy group and 32% in the no-treatment group. However, at 1 year the injection group had only a 69% success rate, compared with 92% for the physiotherapy group and 83% for the wait-and-see group. These differences were statistically significant.

A more recent study by Coombes and colleagues[20] found similar results. They randomized 165 patients in double-blind fashion to CSI or placebo injection, and the two groups were further randomized to physical therapy and bracing or no other treatment. Similar to other studies, at 4 weeks the CSI group was superior to the placebo group in pain relief, but only in the 2 groups that did not have the physical therapy and bracing intervention. The 2 injection groups were no different if the patients also had physical therapy and bracing. At 6 and 12 months, there were inferior results for the groups that had CSIs. For example, at 1 year the placebo injection groups had 96% complete or high-level improvement, compared with 83% in the CSI group. Similarly, at 1 year, the CSI group was much more likely to have a recurrence of the symptoms: 54% versus 12%. This percentage translated into a number needed to harm of 3, in terms of recurrent symptoms.

Corticosteroid Injection for Lateral Epicondylalgia Syndrome: Safety

The only side effects were local skin depigmentation and subcutaneous atrophy, which occurred in about 4% of the CS group and none of the placebo injection group. These changes were mild or moderate and self-limited.

Corticosteroid Injection for Lateral Epicondylalgia Syndrome: Cost

Total cost for CSI for epicondylalgia (medication and physician fee) can vary, but generally ranges from $50 to $200.

Corticosteroid Injection for Lateral Epicondylalgia Syndrome: Conclusions

CSIs for lateral epicondylalgia are equal to but not better than physical therapy at 4 weeks, but carry the longer-term harm of leading to higher recurrence rates and more symptoms at 6 and 12 months. Thus, CSIs for this syndrome should rarely be used, unless there is a compelling reason to offer the patient a short-term relief of symptoms in unusually severe cases.

GREATER TROCHANTERIC PAIN SYNDROME

Greater trochanteric pain syndrome (GTPS), previously termed trochanteric bursitis, is a condition characterized by pain in the lateral hip and thigh region, usually with palpable tenderness in the vicinity of the greater trochanter of the femur. The risk factors for this include obesity and knee osteoarthritis, among other situations that may change the normal biomechanics of gait. The exact pathophysiology is not known, but seems to involve problems in the trochanteric bursa as well as the gluteus medius, although inflammation has not definitively be shown to be involved in this disorder.[21] In many patients, the condition is chronic, and CSIs have been used for decades to treat GTPS, but most of the studies showing benefit to these injections have been methodologically flawed, such as failing to include a placebo group.[22–24]

Corticosteroid Injection for Treatment of Greater Trochanteric Pain Syndrome: Efficacy

Brinks and colleagues[25] randomized 120 patients with GTPS to CSI or usual care (oral analgesia, and physical therapy/exercises as desired by the patients). There was no placebo injection. At 3 months, more of the CSI group recovered compared with the usual-care group (55% versus 34%; NNT = 5). There was also a significant reduction in overall pain in the injection group. However, at 12 months 60% of patients in both groups had recovered, and there were no differences between the two groups in overall pain.

Rompe and colleagues[26] randomized 229 patients with refractory GTPS to 1 of 3 groups: home-training exercise program, a single local CSI, or repetitive low-energy shock wave treatment. At 1 month, the injection group had a much higher success rate (complete recovery or much improvement, 75% vs 7% for the home-training program and 13% for the shock wave therapy). However, at 6 months there was no difference between the CSI group and home-training group (51% and 41% success rates respectively), and at 15 months the shock wave and home-training groups were significantly more successful than the CSI group (80% success for home training vs 48% for the injection group). There were no significant side effects noted in this study.

Corticosteroid Injection for Treatment of Greater Trochanteric Pain Syndrome: Safety

Other than the aforementioned potential worse long-term outcomes, the studies do not report any significant negative consequences to CSI.

Corticosteroid Injection for Treatment of Greater Trochanteric Pain Syndrome: Cost

Total cost for CSI for GTPS (medication and physician fee) can vary, but generally ranges from $50 to $200.

Corticosteroid Injection for Treatment of Greater Trochanteric Pain Syndrome: Conclusion

CSI for treatment of GTPS is probably helpful in the short term, but possibly harmful in the long term (although only in 1 small study). In patients with significant pain, it is reasonable to do CSI for relief of symptoms, in concert with a physical therapy regimen.

DE QUERVAIN TENOSYNOVITIS

De Quervain tenosynovitis, or radial tenosynovitis, is characterized by pain at the thumb side of the wrist, caused by impairment of movement of 2 thumb tendons, the abductor pollicis longus and extensor pollicis brevis. It is often an overuse injury. CSIs have been used to treat this condition, although the studies showing benefit are few.

Corticosteroid Injection for De Quervain Tenosynovitis: Efficacy

In one study,[27] 18 pregnant or lactating women were randomized to CS plus bupivacaine injection or thumb spica splinting. After 6 days, all 9 of the women who had the injection were pain free, whereas none of the splinting group were pain free. No longer-term information was given from this study.

In another study,[28] 21 patients with de Quervain tenosynovitis were randomized to CS or saline injection. At 1 to 2 weeks of follow-up, 75% of the CS group had substantial improvement, compared with only 25% of the placebo group. The steroid responders were then followed for a year, and most, but not all, had maintenance of the improvement. The placebo group had a follow-up CSI as well, so no comparison at 1 year was possible between the two groups.

De Quervain Tenosynovitis: Safety

There were no significant side effects reported in these studies, but no long-term results are available. It is reasonable to assume that there are small potential risks such as tendon rupture and subcutaneous atrophy at the site, which have been reported occasionally.

De Quervain Tenosynovitis: Cost

Total cost for CSI for de Quervain tenosynovitis (medication and physician fee) can vary, but generally ranges from $50 to $200.

De Quervain Tenosynovitis: Conclusion

CSIs for de Quervain tenosynovitis are very helpful, and often curative, in the short term. There is no information to guide clinicians on long-term benefits or risks, but given the significant amount of pain many patients have with this condition it is reasonable to use CSI for primary treatment in patients who understand that there are no long-term data. For patients with lesser amounts of pain, it is reasonable to hold off on CSI and do splinting or other similar interventions.

DISEASES OF THE LUMBAR SPINE

There are different conditions that can affect the lumbar spine, and they differ in symptoms and pathophysiology, although they are related in some ways to degenerative arthritis, disc disease, or both. Sciatica, or lumbar radiculopathy, is a common condition that has a lifetime prevalence of about 4%.[29] It is characterized by radiating lower extremity pain below the knee, with or without low back pain, and is often accompanied by sensory neuropathic symptoms in a dermatomal distribution. Although the prognosis of sciatica is considered favorable, about 30% of patients have persistent symptoms at 1 year, and 5% to 10% proceed to surgery.[30] Similarly, lumbar spinal stenosis is a common condition affecting about 3% of patients, mostly older adults, and is caused by spinal canal narrowing by degenerative changes and congenital causes. This condition is characterized by bilateral radiating symptoms into the legs, and can lead to substantial debilitation. In addition, chronic low back pain is a very common condition that also leads to disability and loss of function.

Treatment of all of these conditions with physical therapy and oral analgesia has significant limitations, and surgery is considered a last resort in most cases. Thus, CSI into the lumbar spine region has emerged as an option in the treatment, and in Medicare recipients alone was used 1.5 million times in 2004,[31] and doubled in number between 1996 and 2007.[32] However, many of the studies have serious methodological flaws and the efficacy of this expensive intervention has been called into question.

Treatment of Lumbar Spine Disease with Corticosteroid Injection: Efficacy

In the WEST study,[32] 228 patients with unilateral sciatica for 1 to 18 months were randomized to 3 epidural injections of CS (triamcinolone) or saline (placebo group). At the 3 weeks, there was a small statistically significant benefit seen in the CSI group, in that 12% of them had greater than 75% improvement in symptoms, compared with 4% of the placebo injection group. Another way of looking at that is that almost all of the patients did not derive a large benefit. There was also a small improvement in overall function in the CS group. However, the benefits seen at 3 weeks disappeared at all time points between 6 weeks and 52 weeks, and the CSIs did not hasten the return to work over time or reduce overall analgesia use. In addition, the two groups did not differ in eventual surgical intervention (15% in both groups).

Another study assessed the utility of epidural CSIs in the treatment of lumbar spinal stenosis.[33] Four-hundred patients with moderate to severe pain thought to be from lumbar spinal stenosis were randomized to epidural injection of CS/lidocaine or lidocaine alone, with a second injection given at 3 weeks at the discretion of the physician. At 6 weeks, there was no significant difference between the two groups, with 38% reporting more than 50% improvement in leg pain and 20% having greater than 50% improvement in disability scores. Both groups had few side effects, with 2% experiencing headache after the procedure, and 3% reporting severe pain from the procedure. Note that 10% of the CS group had adrenal function suppression, compared with none in the lidocaine-alone group, as measured by morning cortisol values. The clinical relevance of this finding is debatable.

Two systematic review and meta-analysis articles were published in the past 4 years, concerning the efficacy of epidural CSIs for radiculopathy and spinal stenosis.[34,35] They similarly concluded, based on about 24 articles each, that epidural CSIs have short-term (a few weeks) benefit in the treatment of radiculopathy/sciatica but not spinal stenosis, and that these benefits are small and of debatable clinical relevance. They also determined that epidural CSIs have no significant benefit on pain relief or disability in the long term, and do not reduce the need for surgical intervention. Both reviews noted that many of the studies did not report harms, but those that did failed to show any substantial harm other than localized discomfort from the procedures themselves.

In addition, a review for the Cochrane group studied the efficacy of CSI for low back pain.[36,37] It concluded that the literature was heterogeneous, because of different locations of CSI (eg, facet joint, epidural, transforaminal), and there was insufficient evidence supporting the use of CSIs for chronic low back pain. The investigators did not rule out a possible benefit for certain subgroups, despite the lack of any solid evidence in support of this.

Treatment of Lumbar Spine Disease with Corticosteroid Injection: Safety

Various short-term mild side effects are reported, such as headache, nausea, and rash, all occurring in 1% to 3% of patients.[33] More serious complications such as dural puncture and infection are rare. Concerns have been raised about an increased risk of future vertebral fracture associated with repeated epidural CSI,[38] but this has not been definitively determined to be a real association. In addition, a much-publicized outbreak of aspergillus meningitis seriously sickened dozens of people and was fatal in several cases. This outbreak was traced back to 1 compounding pharmacy and contaminated methylprednisolone vials.[39]

Treatment of Lumbar Spine Disease with Corticosteroid Injection: Cost

The cost of CSI for lumbar spine disease can vary greatly and depends on the type of procedure used, the use of fluoroscopy, and the use of sedation. With all of these variables, costs can be as high as $2000 to $3000, especially if patients undergo repeat injections over time.

Treatment of Lumbar Spine Disease with Corticosteroid Injection: Conclusions

Epidural CSIs for the treatment of lumbar radiculopathy syndromes and spinal stenosis offer small short-term benefits (weeks) in some patients, but do not show long-term benefits or reduce the need for spinal surgery. The benefits are even more dubious in patients with chronic back pain who undergo facet joint or transforaminal CSI. As such, these interventions should be offered to patients only when a few weeks of pain relief is thought to be worth the cost of the intervention.

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